

Derivation of the Fisher information matrix for poisson pixel localization

$$\theta = (x_0, y_0, z_0)$$

in your paper θ is a 3n vector for n emitters.

this derivation is the $n=1$ case - single emitter's 3 coordinates

I consider $n=1$ emitter case to compare DH-PSF vs. astigmatic PSF localization precision.

Expected signal in pixel k :

$$\mu_k(\theta) = N q_k(\theta), \quad \sum_{k=1}^K q_k(\theta) = 1$$

q_k is pixel integrated so $\sum q_k = 1$

observation model:

$$Y_k \sim \text{Poisson}(\gamma_k(\theta)), \quad \gamma_k(\theta) = \mu_k(\theta) + b$$

- ✓ each pixel is an independent poisson random variable
- ✓ photon counting noise is physically poisson. b is background and does not depend on θ .

joint probability mass function:

$$P(Y|\theta) = \prod_{k=1}^K \frac{e^{-\gamma_k(\theta)} \gamma_k(\theta)^{Y_k}}{Y_k!}$$

Log-likelihood:

$$l(\theta) = \sum_{k=1}^K [Y_k \ln \gamma_k(\theta) - \gamma_k(\theta) - \ln(Y_k!)]$$

$$\frac{\partial l}{\partial \theta_i} = \sum_{k=1}^K \left(\frac{Y_k}{\gamma_k(\theta)} - 1 \right) \frac{\partial \gamma_k(\theta)}{\partial \theta_i}$$

$$\rightarrow \frac{\partial}{\partial \theta_i} [Y_k \ln \gamma_k] = \cancel{Y_k \ln \gamma_k} \frac{Y_k}{\gamma_k} \frac{\partial \gamma_k}{\partial \theta_i}$$

$$\rightarrow \frac{\partial}{\partial \theta_i} [-\gamma_k] = -\frac{\partial \gamma_k}{\partial \theta_i}$$

Definition of the FIM:

$$I_{ij}(\theta) = E \left[\frac{\partial l}{\partial \theta_i} \frac{\partial l}{\partial \theta_j} \right] = - E \left[\frac{\partial^2 l}{\partial \theta_i \partial \theta_j} \right]$$

now the second derivative of l :

$$\frac{\partial^2 l}{\partial \theta_i \partial \theta_j} = \sum_{k=1}^K \left[-\frac{Y_k}{\gamma_k^2} \frac{\partial \gamma_k}{\partial \theta_i} \frac{\partial \gamma_k}{\partial \theta_j} + \left(\frac{Y_k}{\gamma_k} - 1 \right) \frac{\partial^2 \gamma_k}{\partial \theta_i \partial \theta_j} \right]$$

~~proof of the~~

key poisson identity:

$$E[Y_k] = \lambda_k(\theta)$$

$$E\left[\frac{Y_k}{\lambda_k}\right] = 1 \Rightarrow E\left[\frac{Y_k}{\lambda_k} - 1\right] = 0$$

This makes the second term vanish under expectation:

$$E\left[\frac{\partial^2 \ell}{\partial \theta_i \partial \theta_j}\right] = - \sum_{k=1}^K \frac{1}{\lambda_k(\theta)} \frac{\partial \lambda_k}{\partial \theta_i} \frac{\partial \lambda_k}{\partial \theta_j} \quad (*)$$

the second derivative term disappears in expectation because the poisson mean equals the poisson rate.

considering $I_{ij} = -E\left[\frac{\partial^2 \ell}{\partial \theta_i \partial \theta_j}\right]$ and $(*)$

$$I_{ij}(\theta) = \sum_{k=1}^K \frac{1}{\lambda_k(\theta)} \frac{\partial \lambda_k(\theta)}{\partial \theta_i} \frac{\partial \lambda_k(\theta)}{\partial \theta_j}$$

since $\lambda_k = N_{g_k}(\theta) + b$

$$\frac{\partial \lambda_k}{\partial \theta_i} = N \frac{\partial g_k}{\partial \theta_i}$$

substituting

$$I_{ij}(\theta) = N^2 \sum_{k=1}^K \frac{1}{N_{g_k}(\theta) + b} \frac{\partial g_k}{\partial \theta_i} \frac{\partial g_k}{\partial \theta_j}$$

$g_k(\theta)$ is the gradient of the pSF at pixel k :

$$g_k(\theta) = \nabla_{\theta} q_k(\theta) \in \mathbb{R}^3$$

$$\rightarrow I(\theta) = N^2 \sum_{k=1}^K \frac{g_k(\theta) g_k(\theta)^T}{N q_k(\theta) \leftarrow b}$$

Cramer-Rao Lower Bound:

for any unbiased estimator:

$$\text{Cor}(\hat{\theta}) \geq I(\theta)^{-1}$$

$$\hookrightarrow \sigma_x = \sqrt{[I(\theta)^{-1}]_{11}}$$

$$\sigma_y = \sqrt{[I(\theta)^{-1}]_{22}}$$

$$\sigma_{22} = \sqrt{[I(\theta)^{-1}]_{33}}$$

✓ Fisher information measures how sharply peaked the likelihood is around the true parameter. more information means a lower achievable variance.

result:

using

$$I(\theta) = N^2 \sum_{k=1}^K \frac{g_k(\theta) g_k(\theta)^T}{N g_k(\theta) + b}, \quad \sigma_i = \sqrt{[I(\theta)^{-1}]_{ii}}$$

with $N = 3000$ photons, $B\text{-eff} = 0.0125$ photons/pixel

, 25×25 pixel patch

the Fisher information matrix at the focal plane ($z=0$)

for Astigmatic PSF:

$$I(\theta) = \begin{pmatrix} 0.1415 & 0 & 0 \\ 0 & 0.1515 & 0 \\ 0 & 0 & 0.000408 \end{pmatrix} \text{ nm}^{-2}$$

$$\rightarrow \sigma_x = 2.658 \text{ nm}$$

$$\sigma_y = 2.570 \text{ nm}$$

$$\sigma_z = 49.496 \text{ nm}$$

for Double-helix PSF:

$$I(\theta) = \begin{pmatrix} 0.1513 & 0 & 0 \\ 0 & 0.2052 & 0 \\ 0 & 0 & 0.3025 \end{pmatrix} \text{ nm}^{-2}$$

$$\sigma_x = 2.571 \text{ nm}$$

$$\sigma_y = 2.208 \text{ nm}$$

$$\sigma_z = 1.818 \text{ nm}$$

✓ at focus the astigmatic PSF I_{zz} is nearly zero so the PSF shape barely change with z , producing a $27\times$ worse axial CRLB than DH-PSF.